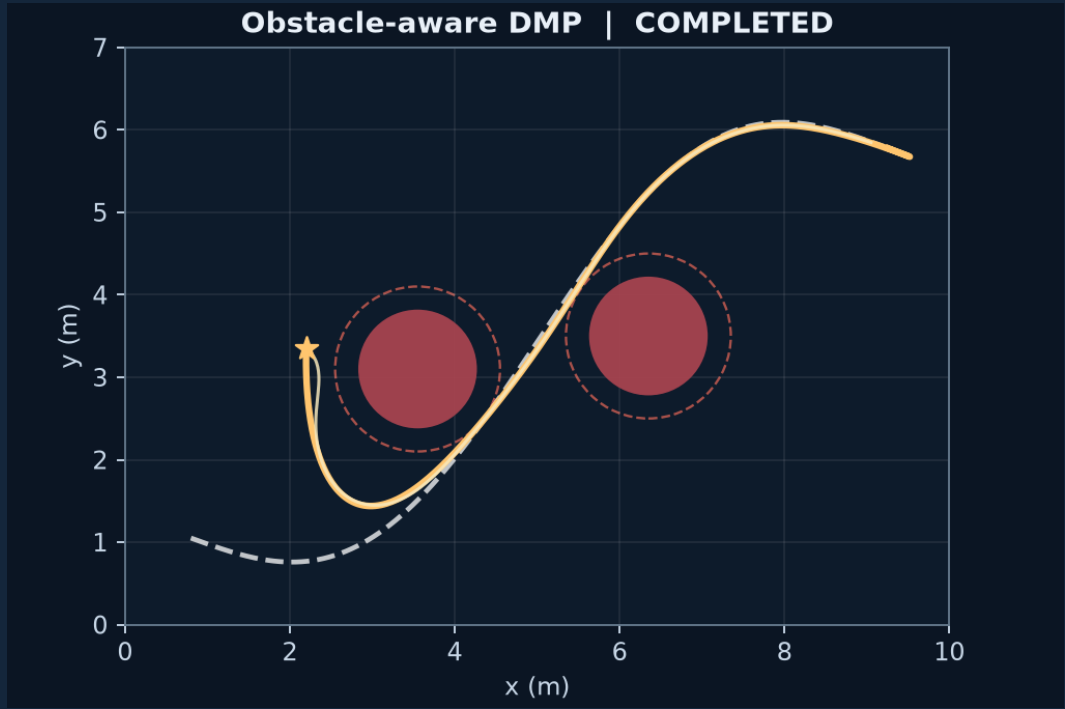
Teach & Repeat Rover Navigation

Adapting one demonstrated route from a displaced start

FOUR METHODS • 60 START POSES EACH

Ayush Kaulgud · Irenov Joel · Tapisar Singh



Obstacle-aware DMP / successful example

The question and our current prototype

Can the rover safely rejoin a taught route when it starts elsewhere?

Same route. Same map. Same feedback controller. Only the recovery trajectory changes.

INITIAL SIMULATION / 2-D STATIC MAP

01	**Teach**	Record a safe reference route
02	**ICP prototype**	Synthetic registration, separate from trials
03	**Recover**	Generate one of four paths
04	**Check**	Reject unsafe plans; monitor margin
05	**Repeat**	Follow with a shared controller

Four recovery methods, one test harness

01 Direct waypoints

Connect to the nearest forward taught point, then follow stored waypoints.

02 Local reconnection

Use grid A* to find a collision-free merge onto the demonstrated route.

03 Classical DMP

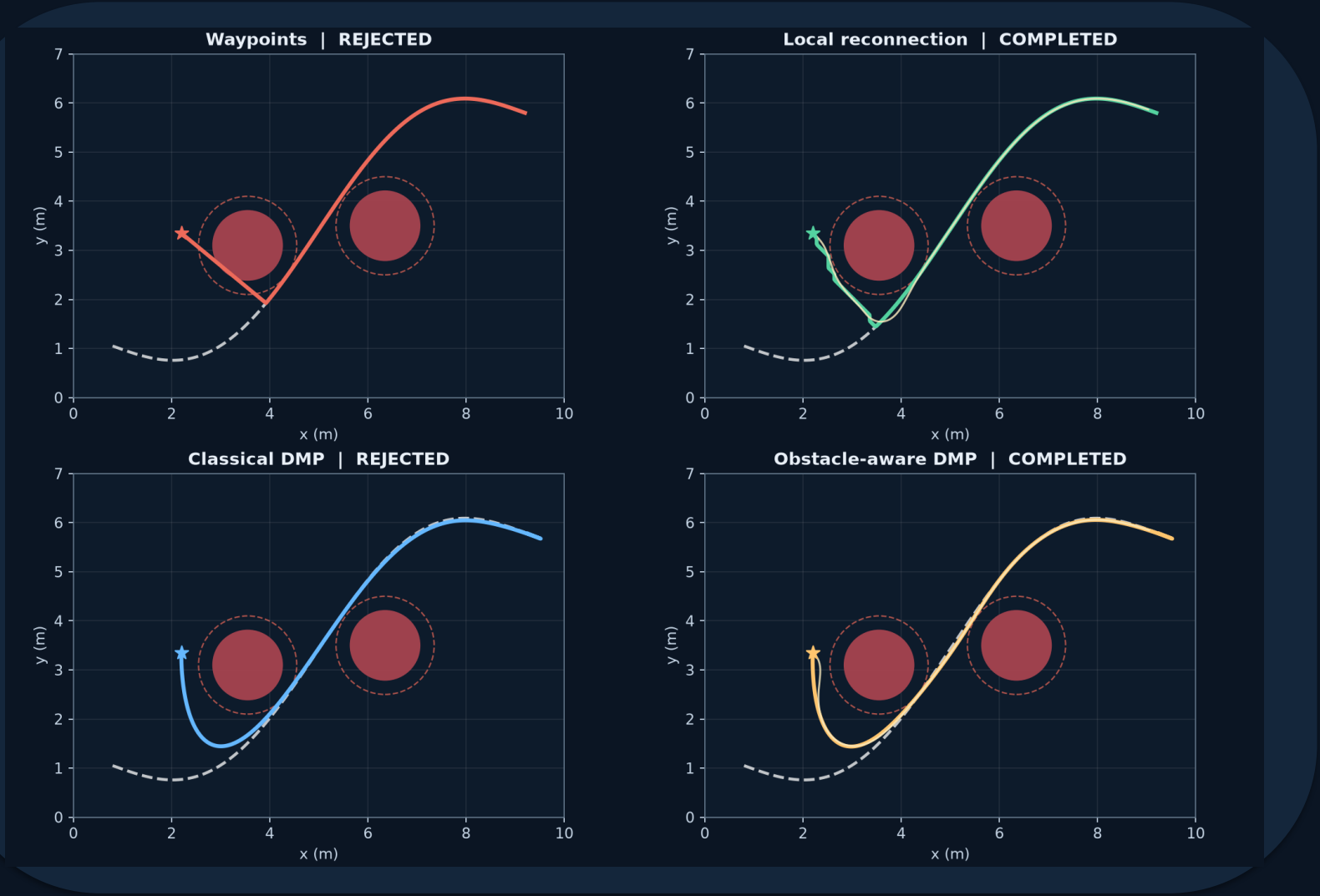
Fit a discrete DMP to the demonstration; regenerate from the new start.

04 Obstacle-aware DMP

Add a repulsive obstacle coupling term during DMP rollout.

ICP is the shared localization component; it is not a fifth trajectory planner.

One displaced start, four visibly different outcomes



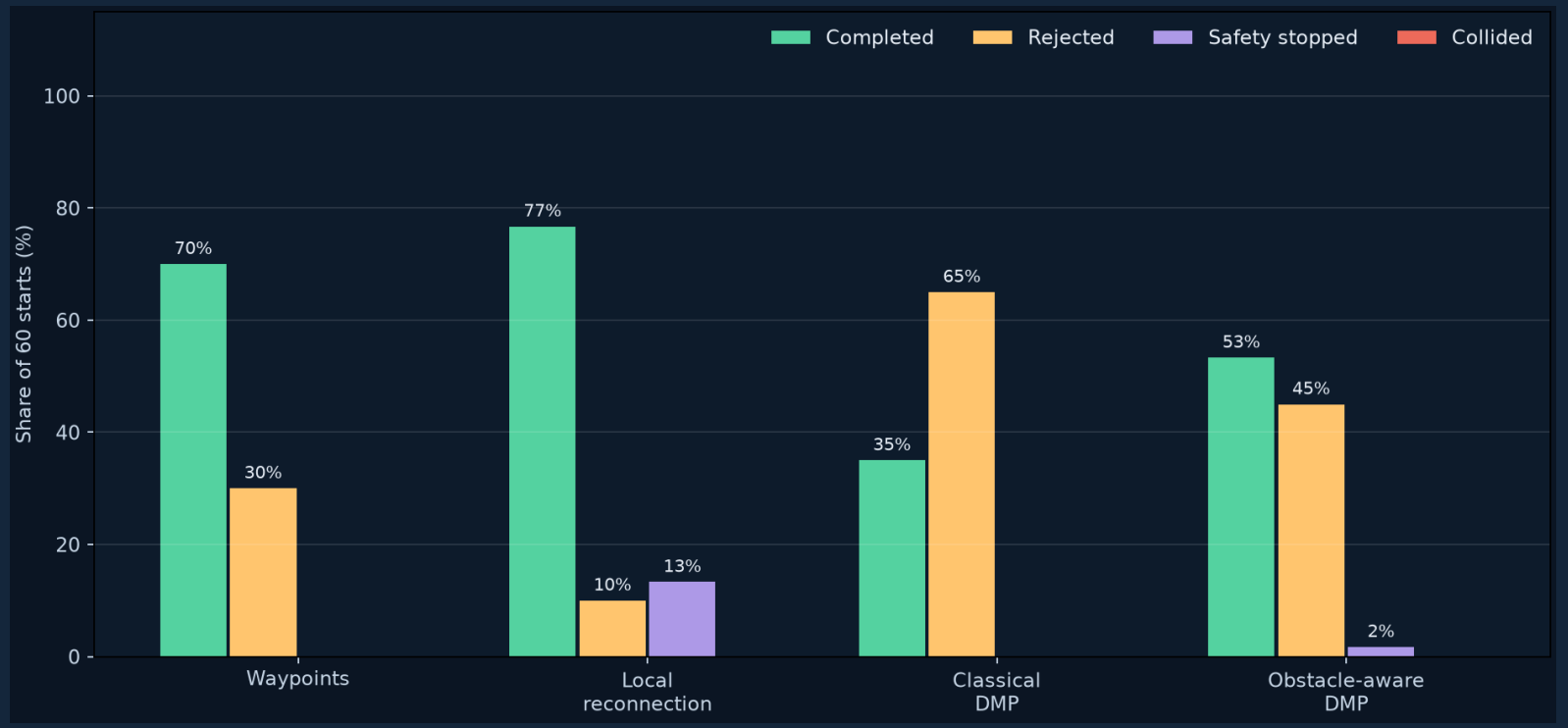
EXAMPLE START

(2.2, 3.35) m
−35° heading

Local reconnection and obstacle-aware DMP completed this run.

Dashed white = taught route
Colored = planned path
Cream = rover trace
Red rings = safety buffer

Initial outcome across 60 starts per method



76.7%

Highest completion: A*

+18.3 pp

Coupling vs. classical DMP

0%

Observed collision rate

Safety stops are counted separately from path rejections.

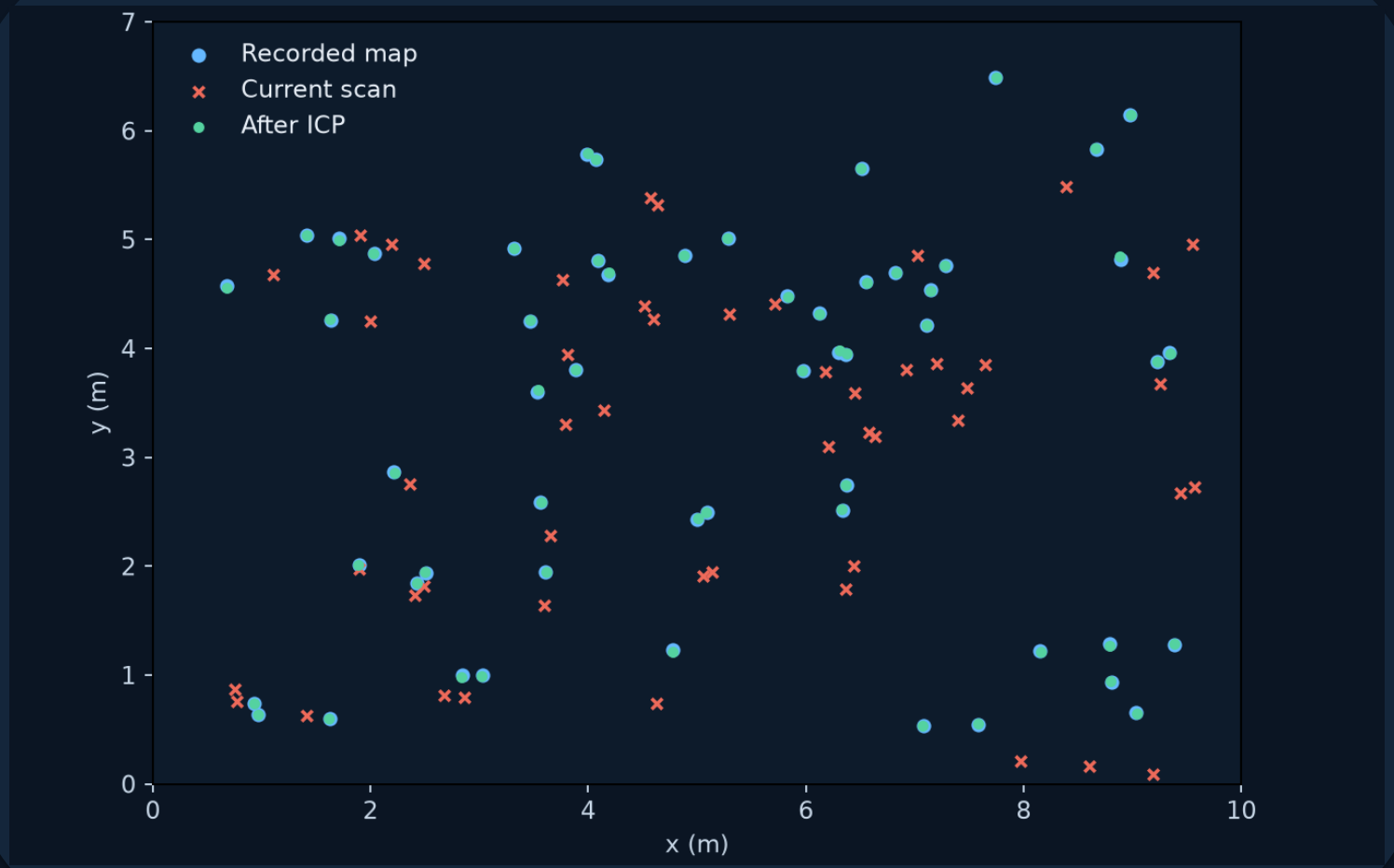
Metrics from the original project plan

METHOD	COMPLETE	REJECT	STOP	MIN CLR	GOAL ERR	PLAN TIME
Waypoints	70.0%	30.0%	0.0%	0.130 m	0.169 m	0.05 ms
Local reconnection	76.7%	10.0%	13.3%	0.108 m	0.168 m	2.70 ms
Classical DMP	35.0%	65.0%	0.0%	0.122 m	0.163 m	3.12 ms
Obstacle-aware DMP	53.3%	45.0%	1.7%	0.119 m	0.165 m	4.33 ms

Minimum clearance is measured on executed traces. Goal error is the median for completed runs.

Planning time depends on this laptop and is descriptive, not a hardware benchmark.

ICP localization proof of concept



0.012 m

Registration RMSE

0.0016 m

Translation error

0.008°

Rotation error

Synthetic 55-point scan; ICP is demonstrated separately from navigation trials.

What we learned and what comes next

EARLY FINDINGS

01 A* reconnection completed the most starts in this map.

02 Obstacle coupling improved DMP completion by 18.3 points.

03 Runtime safety stops exposed controller corner-cutting.

NEXT MILESTONES

01 Feed periodic ICP corrections into noisy odometry runs.

02 Test more obstacle layouts and larger pose offsets.

03 Tune controller and DMP coupling; report uncertainty.

Experiment details and reproducibility

TRIAL DESIGN

- 60 starts = 4 x positions × 5 y positions × 3 headings
- 10 × 7 m static map with two circular obstacles
- Rover radius 0.16 m; path margin 0.12 m
- Deterministic bounded unicycle controller
- Unsafe paths rejected before execution; margin monitored online

REPRODUCE

```
python src/robot_motion.py
python build_slides.py
```

The script exports trial-level CSV, summary JSON, and the figures shown here.

Sources: class proposal and literature review; Ijspeert et al. (2013), Hoffmann et al. (2009), Besl & McKay (1992).